

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I _Narmatha J (192521341)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. CLOUD COMPUTING:

Cloud Computing is the delivery of computing services such as servers, resources, networking, providers, storage over the internet on demand.

CHARACTERISTICS:

*. On-demand Self-service:

Users can access resources whenever needed without human intervention.

*. Broad Network Access:

Servers are available over the internet from different devices.

*. Rapid Elasticity: Resources can be increased or decreased quickly based on the demands.

*. Resource pooling: Allows multiple users to access or using virtualization.

2. EVOLUTION OF CLOUD COMPUTING:

- *. Mainframe Computers: Users access resources from centralised systems.
- *. Personal Computers: Users can access their own computers using servers.
- *. Client-Server Computers: Client requested the resources from the centralised servers.
- *. Virtualization: Allows multiple virtual machines to run on a single physical server to improve computing utilization.
- *. Cloud Computing: Services are delivered to the resources ^{such as IaaS, PaaS and SaaS} over the internet.

3. ROLE OF SERVER VIRTUALIZATION:

- *. Allows multiple virtual machines run on a one physical server.
- *. It improves computing utilization.
- *. Reduce cost and energy efficiency.
- *. It is more management and scalability and helps in backup and recovery.

VIRTUALIZATION	CLOUD COMPUTING
*. Creates multiple virtual machines to provide resources.	*. Service models deliver the resource to users.
*. Focuses on efficient on hardware usage.	*. Focuses on resources to be delivered.
*. Can exist without cloud computing.	*. Virtualisation is one of its technologies.

4. ROLE OF DATA CENTERS :

- *. Store data, storage devices, networking devices
- *. Uses firewalls and backups for recovery.
- *. Host web application and web services.
- *. Has high scalability to enhance the correction of data through services.
- *. It has the facility to store data in large amounts to access it later.
- *. It has high reliability and scalability.

5. ⇒ IaaS (Industry as a Service) :

*. It provides the servers, to the user to make the applications.

Eg: Amazon ECS

⇒ PaaS (Platform as a Service) :

*. It provides the platform to develop and deploy the application.

Eg: Google App Engine.

⇒ SaaS (Software as a Service) :

*. It provides ready-to-use software over the internet.

Eg: Google docs.

⇒ XaaS (Anything as a Service) :

*. It provides everything to the user needs over the internet to build the application.

Eg: Microsoft 365